

Literature Review Draft (April 2025)
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for Filipa Calado

INTRODUCTION

The purpose of this literature review is to survey how gender bias in text is analyzed, with a focus on developing a schema which will analyze transphobia in LLM-generated texts. This undertaking begins by looking at common frameworks of evaluating gender bias in LLMs, and critiquing their binary categories from a framework of trans studies. In response Kadji Amin's framework of transition and identification is introduced and analyzed through a Marxist lens, then connected back to the current transgender 'cultural crises', as well as the historical one which precipitated the establishment of Nazi Germany. Amin's theories of transition and identification are then evaluated based on their relevance to a potential schema, as well as to the binary categories discussed earlier. A discussion is had on dividing versus linking instances and outcomes of bias, with a preference established for linking in this case. Hannah Devinney's use of non-exclusive keyword tagging for evaluating how LLM bias is analyzed in academia, is considered as a model for evaluating LLM bias itself. From this a draft schema of keywords, their definitions, and how they may appear in language generated from anti-trans legislation is proposed, with potential next steps moving forward.

EXISTING FRAMEWORKS

The start of my process was reviewing Filipa Calado "Some Myths About Bias" (2025), which analyzes existing frameworks for categorizing gender bias within LLM-generated text. Calado looks at three texts in particular, "Proposed Taxonomy for Gender Bias in Text" (Hitti et al. 2019), "Gender bias in transformers" (Nemani et al. 2023), and "The Problem with Bias" (Barocas et al. 2017).

Hitti et al. (2019) categorize gender bias in language as structural or contextual (definitions in Table 1). They specify that structural harm only occurs when gender assumptions are enforced in a *gender neutral* setting (i.e. an LLM assuming a doctor uses he/him is structural, but an LLM assuming a pregnant doctor uses she/her is not, because 'pregnancy' is indicative of a female subject).

Nemani et al. (2023) categorizes gender bias as either denigration, stereotyping, or underrepresentation (definitions in Table 1). Underrepresentation is commonly discussed as a perpetrator and result of gender bias in language, but no papers reviewed discussed over-representation, or the repercussions of hyper-visibility. Stereotypes are also frequently discussed in the literature, but only in a yes/no manner (are there vs. aren't there stereotypes) rather than the content or assigned value (positive vs. negative) of the stereotypes.

Barocas et al. (2017) categorizes gender bias as producing either allocative harms or representational harms (definitions in Table 1).

Table 1: Definitions from Hitti et al., Nemani et al., and

Hitti et al.	contextual harm	bias which re-iterates social and behavioral stereotypes around gender roles
	structural harm	“structural gender bias occurs when bias can be traced down from a specific grammatical construction. This includes looking up of any syntactic patterns or keywords that enforce gender assumptions in a gender neutral setting”
Nemani et al.	denigration	“derogatory language aimed at demeaning specific groups”
	stereotyping	“individuals’ [or machines’] cognitive representation of a particular social group, where certain biases are learned and perpetuated”
	under-representation	“absence of identifiable group members from representative bodies and well-being indicators in proportion to their population’s size”
Barocas et al.	allocative harm	language which results in resources being withheld from certain groups
	representational harm	language which results in certain groups being under-represented or stereotyped

CRITIQUE

In Calado 2025 the use of binary schemas for analyzing gender bias in language is critiqued. The use of either/or, zero-sum categories is also a frequent critique of the male/female binary, exemplified in Judith Butler’s *Gender Troubles* (1990). Ironically, since its publication *Gender Troubles* has frequently been misread as simplistically proposing a ‘replacement’ binary of sex (biology “at birth”) versus gender (identity). This concept is not only prevalent within mainstream ideas of transexuality and trans-affirming activisms, but also in almost all current anti-transgender legislation being proposed on the federal level (<https://translegislation.com/>). More recently some prominent scholars within trans studies have instead theorized gender roles through universal and/or overlapping frameworks, such as Angelea Long Chu in *Females* (2019) and Kadji Amin in “We Are All Nonbinary” (2022).

TRANSITION/IDENTIFICATION

In “We Are All Nonbinary” Amin offers an alternative framework from that of assigned sex at birth vs. gender identity. Taking a trans materialist perspective, Amin describes ‘transness’ as encompassing two categories: transition and identification. Transition is “a shift in social gender categories, whatever they may be”. According to Marxism, transition is a materialist process, an idea furthered by Jules Gill-Peterson who defines transitioning as “less a volitional act of identification, let alone a subversion of the gender binary, than a concrete response to living and working in the world” in the Introduction to Transgender Studies Quarterly (TSQ)’s *Trans Marxism* issue (2024).

Amin describes identification as “the psychic process that makes the interval between the individual and the social apparent”. Rather than an either/or binary, Amin theorizes these concepts as interrelated. Identification is not the “site of suture” (as is conceptualized in a sex-at-birth/gender-identity binary), but is a psychological process of making the “site of the suture between the individual and the social” (i.e. transition, in whatever form) apparent. Identification is a process created by those transitioning as a means of making their experiences socially legible within an individualist, postmodernist framework. According to Marxism, identification would be the fetishistic projection of transition, akin to commodity fetishism, which alienates the process of transition from the material conditions it arises from, allowing it legibility and sanction (at least partly) within industrial capitalism.

Table 2: Definitions from Amin (2022)

transition	“a shift in social gender categories, whatever they may be”
identification	“the psychic process that makes the interval between the individual and the social [i.e. transition] apparent”

TRANSITION AND CAPITALISM

These connections between hypercapitalism and the process of identification are outlined in Jonah Peretti’s 1996 essay “Capitalism and Schizophrenia” (published a decade before he founded BuzzFeed). Peretti posits that “identity formation is inextricably linked to the urge to consume, and therefore the acceleration of capitalism necessitates an increase in the rate at which individuals assume and shed identities. The internet is one of many late capitalist phenomena that allow for more flexible, rapid, and profitable mechanisms of identity formation”. Interestingly Peretti’s essay goes on to name Judith Butler’s practice, specifically her “queer political performance”, as a radical means of weaponizing production against the machineries of capitalism. Peretti sees this as the opposite of identity-centered queer

activism, which functions to further integrate and assimilate queer people into capitalism.

The TSQ's *Trans-Marxism* (2024) offers a deeper (and more contemporary) perspective on the subject. Wren Ariel Gould's essay "Coconstructing (Trans-)Capitalist Realism" explores the latent support of late capitalism by trans-affirming activism, particularly when responding to the recent trend of anti-trans activism. Gould proposes that trans-affirming activism works to perpetuate capitalism by obscuring the purpose of anti-trans legislation. Rather than being primarily 'identity' motivated, these policies actually exist to garner popular support for austerity policies aiming to destroy public institutions (the ACA, public LAMs, and public schools & universities), which would otherwise be viewed as authoritarian measures.

In addition to providing popular support for unpopular policies, Gould also asserts that a rhetorical focus on trans people is an inevitable result of late capitalism. "Late capitalism necessarily and materially foregrounds transgender 'cultural crises'" because "transgender subjects are attached to late capitalism, through their roles as consumers (of medicalized and nonmedicalized technologies of the body) and as workers (who either endorse late capitalism via assimilation or constitute... a surplus work force relegated to sex work that supports global sex trade as well as disciplining the global proletariat)" (2024).

It is perhaps of interest that these connections between transsexuality and the progression of industrial capitalism also echo the conditions within the Weimar Republic and its eventual transition to the Third Reich. The first book burning by the Nazi Party targeted the libraries, archives, and dormitories of the Institut für Sexualwissenschaft, commonly associated with Magnus Hirschfeld. The world's largest collection of materials on gender and sexuality was destroyed, and the majority of residents- patients receiving HRT and/or GAS, housed and employed at the Institut with menial jobs- were murdered.

This trend towards anti-trans laws and violence following industrial revolutions (such as the Information Age), could reflect an instinct/aim to preserve essentialist gender roles and associations. As technology exponentially innovates so too do the technologies of gender and sex, opening the possibility for greater recognition of them as constructs mediated through technologies rather than essential features granted at, or before, birth (theorized further in Paul Preciado's *Testo Junkie* (2013)). By creating trans cultural crises, this focus can be redirected towards naturalizing gender for those who haven't transitioned by deeming technological innovations as only being such when they are used for the process of transition.

TRANSITION/IDENTIFICATION AS A SCHEMA

Going back to Amin, I then considered how the concepts of transition and identification could be used in a schema analyzing transphobic language. Creating another 'zero-sum' schema of transition versus identification didn't seem helpful,

especially given Amin’s construction of them as related and interlinked. Also while Amin is critical specifically of identification without transition as a “phantasmic and labile relation to [gender] categories” which “costs very little” (2017), current legislation doesn’t make the same distinctions. Instead contemporary anti-trans rhetoric promotes the idea that identification is a ‘gateway’ to transition to promote alarmist narratives (exemplified by *Irreversible Damage* by Abigail Shrier (2020)). Additionally for our purposes, over-focusing on the binary/non-binary dichotomy of transness has the potential to redirect attention towards overemphasizing identity, rather than the authoritarian aims behind anti-trans policy.

Before going any further I mapped a schema of how the seven categories of bias related to identification and/or transition in terms of their foci. I also mapped them to three categories of direct ‘results’: unequal distribution of resources, creating/reinforcing negative prejudice, and stereotypical and/or missing representation.

Table 3: Schema of terms thus-far

	unequal distributio n	transition (Amin)	negative prejudice	identification (Amin)	representation
H. et al			contextual harm		structural harm
N. et al		denigration		stereotyping	underrepresentation
B. et al	allocative harm		representational harm		

Two potential points of interest from this table is the current literature on gender bias in LLMs clearly focusing on issues of underrepresented and stereotypical depictions rather than negative or directly harmful language, and a lack of attention to how language bias relates to unequal allocation of resources. The overlap between rows, columns, and cells further proves the inherent interrelation and lack of mutual exclusivity between categories of bias which are being posited as such.

DIVIDING VS. LINKING BIAS

Given the correlations and overlaps within categories and effects of bias, a binary or zero-sum schema seems counterproductive to our goal. Instead a schema akin to keyword tagging may be more helpful, allowing bias to be given any number of descriptors- even those which may be commonly represented as mutually exclusive, such as male and female. Ideally this will help avoid ‘dividing bias’ into distinct occurrences, allowing instances and methods of bias to be linked and mapped onto one another, better reflecting how bias operates and revealing hidden connections. Linking biases to one another also follows the intersectional

frameworks of transphobia advocated by Gould (2024) and Butler in *Who's Afraid of Gender?* (2024). This should also help expose the source of anti-trans activism as a means of perpetuating austerity politics and racial capitalism.

KEYWORD TAGGING

This method of keyword tagging to analyze bias is also used in Devinney et al.'s "Theories of "Gender" in NLP Bias Research" (2022). Rather than analyzing bias text generated by LLMs, this paper analyzes bias within almost 200 academic papers written about gender bias in LLMs. Devinney uses six categories: binary, essentialist, neutral, non-binary inclusive, social construct, and trans-inclusive (definitions in Table 4).

Table 4: Keywords from Devinney et al. 2022

binary	considers only men and women
essentialist	makes specific claims about fundamental difference re: gender and/or sex
neutral	includes a concept of 'neutrality' (mixed groups and/or unknowns)
non-binary inclusive	thinks & operationalizes beyond two genders
social construct	acknowledges gender varies given cultural contexts (is a social construction)
trans-inclusive	thinks & operationalizes beyond cis men and women

Building off of these keywords and frameworks of anti-trans laws from Gould (2024) and Butler (2024), I started an early draft of keywords for tagging our future LLM-generated text, including how they may appear within the actual text (Table 5).

Table 5: Draft of schema (in progress)

keyword	definition	bill terms
austerity	aims to limit/restrict funding for public services, namely the ACA, prisons and institutions, public housing, and	public institutions; schools; universities; libraries; funding/federal funds; DEI; housing; hospitals

	public schools and libraries; uses anti-trans rhetoric and legislation to garner popular support for authoritarian austerity measures which would otherwise face mainstream resistance	
essentialist	makes specific claims about fundamental difference re: gender and/or sex	sperm; eggs; XX; XY; at birth; breast-feeding; reproductive capacity; immutable
identification	"the psychic process that makes the interval between the individual and the social [i.e. transition] apparent"	DEI; federal funds; parents; other than assigned; X identification; gender identity
institutionalization	targets individuals unwillingly* in government care (in prison/jail, ICE facilities, psych wards)	prison; ICE; immigration;
medical control/healthcare	restrictions of both medical transition and all healthcare (not just transition) for trans/poc/rural/poor people	surgery; healthcare; doctors; hospitals; mutilation; experiments
protection of children/family	[add]	schools; parents; marriage; children; protection; mutilation
protection of (cis) women	both from trans women, and from potentially transitioning to men	women's spaces; bathrooms; breast-feeding; reproductive capacity; sports; protection
representation	in public; over-; under-	DEI; [includes restrictions on medical transition?]
social construct	acknowledges gender varies given cultural	gender identity; self-perception;

	contexts (is a social construction)	psychological
transfemme	targets trans women	sex-segregation; bathroom; sports; housing; women's spaces; prisons; protecting; public spaces
transmasc	targets trans men	breast feeding; reproductive capacity; adolescent healthcare
transition	"a shift in social gender categories, whatever they may be"	transition; healthcare; surgery; hormone treatment; mutilation; experimentation; sex at birth

*could be expanded to include members of the military

NEXT STEPS (CONCLUSION)

The next step is obviously refining the above schema and incorporating feedback from my collaborators on this project. It could also be interesting to map the above keywords and bill terms to visualize how they might relate to one another. This could offer a small-scale model of what terms might predict other terms in a LLM.